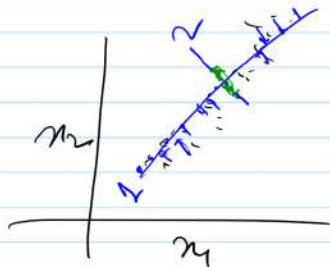


COL774 / COL7341
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PCA := Principal Component Analysis



$$\{x^{(i)}\}_{i=1}^m$$

$$x^{(i)} \in \mathbb{R}^n$$

Principal components

$$u_1, \dots, u_k \quad k \leq n$$

$$u_k \in \mathbb{R}^n$$

$$z_1^{(i)} = x^{(i)T} u_1$$

$$z_2^{(i)} = x^{(i)T} u_2$$

$$x^{(i)} = z_1^{(i)} u_1 + z_2^{(i)} u_2 + \dots$$

Pre-processing step (Normalization)

$$x_j^{(i)} \leftarrow \frac{x_j^{(i)} - \mu_j}{\sigma_j}$$

$$z_k^{(i)} = x^{(i)T} u_k$$

$$u_1, \dots, u_k$$

$$\Rightarrow \text{Projected Variance} = \sum_{k=1}^K \frac{1}{m} \sum_{i=1}^m [x^{(i)T} u_k]^2$$

$$\frac{1}{m} \sum_i x^{(i)T} u_k = 0$$

(Because of pre-processing)

$$\begin{aligned} \Rightarrow \sum_{k=1}^K \frac{1}{m} \sum_{i=1}^m (u_k^T x^{(i)}) (x^{(i)T} u_k) \\ = \sum_{k=1}^K u_k^T \left[\frac{1}{m} \sum_{i=1}^m x^{(i)} x^{(i)T} \right] u_k \\ \Rightarrow \sum_{k=1}^K u_k^T \left[\frac{1}{m} \sum_{i=1}^m x^{(i)} x^{(i)T} \right] u_k \\ \Rightarrow \sum_{k=1}^K u_k^T \left[\frac{1}{m} \sum_{i=1}^m x^{(i)} x^{(i)T} \right] u_k \end{aligned}$$

Empirical co-variance of feature x_j & x_k

$$\Rightarrow u_1, \dots, u_k \text{ s.t.}$$

$$\text{Projected Variance} = \sum_{k=1}^K u_k^T \sum_{i=1}^m u_k$$

s.t. p.v. is maximized

$$\text{Let } k=1 \text{ s.t.}$$

$$\max_{u_1} u_1^T \sum_{i=1}^m u_i = \max_{u_1} - (u_1^T \sum_{i=1}^m u_i)$$

$$\text{s.t. } u_1^T u_1 = 1$$

$$L(u_1, \lambda) = -u_1^T \sum_{i=1}^m u_i + \lambda (u_1^T u_1 - 1)$$

$$\nabla_{u_1} L(u_1, \lambda) = 0 \text{ at optimal point}$$

$$\Rightarrow -2 \sum_{i=1}^n \mu_i + 2\lambda \mu_1 = 0$$

$$\Rightarrow \boxed{\sum_{i=1}^n \mu_i = \lambda \mu_1} \quad (1)$$

$$\Rightarrow \text{Objective: } - \max_{\mu_1} \underbrace{a_1^T \sum_{i=1}^n \mu_i}_{\lambda \mu_1}$$

μ_1 is eigenvector corresponding to highest eigenvalue $\lambda \equiv \lambda^k$:- highest eigenvalue of $\sum_{i=1}^n$

finding μ_2 :-

$$\max_{\mu_1, \mu_2} a_1^T \sum_{i=1}^n \mu_i + a_2^T \sum_{i=1}^n \mu_i$$

μ_1 :- eigenvector corresponding to highest eigenvalue

$$\begin{cases} \mu_1^T \mu_1 = 1 & \mu_2^T \mu_2 = 1 \\ \mu_1^T \mu_2 = 0 \end{cases}$$

$$\mu_1^T \mu_2 = 0$$

$$L(\mu_1, \mu_2, \lambda_1, \lambda_2, \beta) = - \left[a_1^T \sum_{i=1}^n \mu_i + a_2^T \sum_{i=1}^n \mu_i \right]$$

$$\nabla_{\mu_2} L(\mu_1, \mu_2, \lambda_1, \lambda_2, \beta)$$

$$= -2 \sum_{i=1}^n \mu_i + 2\lambda_1 \mu_1 + \beta \mu_1 = 0 \quad (1)$$

$$\nabla_{\mu_1} L(\mu_1, \mu_2, \lambda_1, \lambda_2, \beta)$$

$$= -2 \sum_{i=1}^n \mu_i + 2\lambda_1 \mu_1 + \beta \mu_2 = 0 \quad (2)$$

$\therefore \mu_1, \dots, \mu_k$ are eigenvectors corresponding to highest eigenvalues of $\sum_{i=1}^n$